

Fully Homomorphic Encryption: definitional issues and open problems

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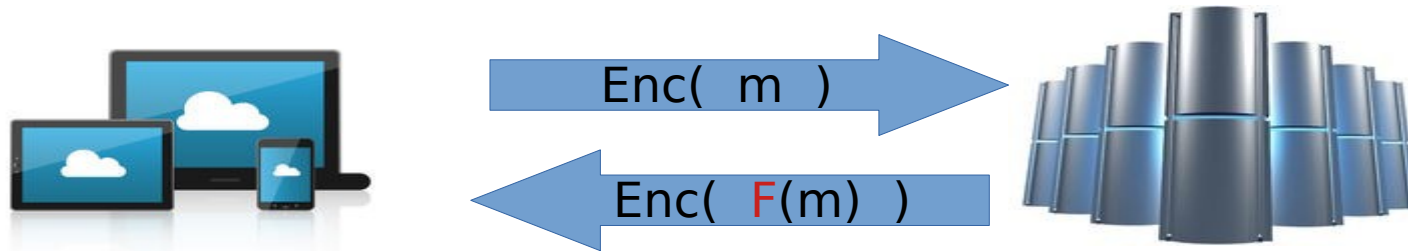


Fully Homomorphic Encryption

- Encryption: used to protect data at rest or in transit



- Fully Homomorphic Encryption: supports arbitrary computations (F) on encrypted data



FHE Timeline / Taxonomy

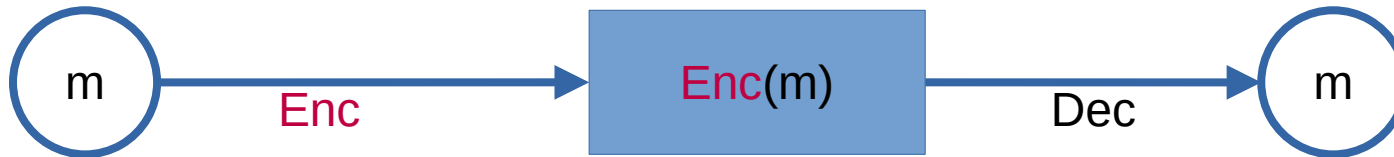
- 1978 [Rivest, Adleman, Dertouzos]: pose problem
- 2009 [Gentry]: **first** candidate solution, **bootstrapping** technique
- 2011 [Brakerski, Vaikuntanathan]
 - **first** solution based on standard lattice problems
- Many other FHE schemes proposed since 2011
 - All based on lattices and use **bootstrapping**
- Usually grouped in “generations”:
 - First generation: BGV, BFV ... **CKKS?**
 - Second generation: GSW ...
 - Third generation: **FHEW**, TFHE ...

Today

- Fully Homomorphic Encryption
 - Traditional definition
- Lattice-based encryption
 - Noise growth and bootstrapping
- Fully Composable Homomorphic Encryption
 - with proofs!
- Approximate encryption (time permitting)

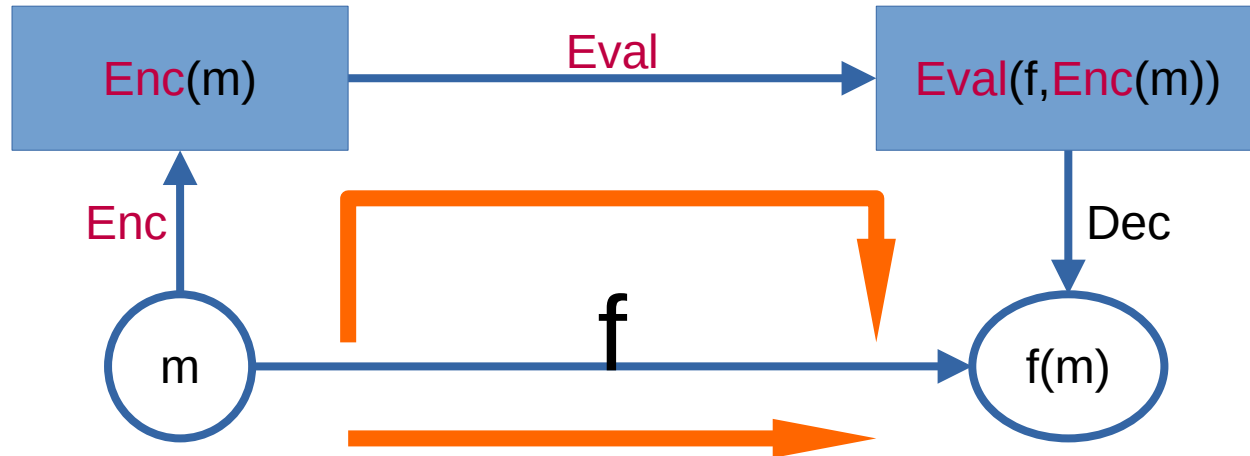
Encryption Scheme

- Syntax: (Gen, **Enc**, Dec)
- Correctness:
 - $(pk, sk) \leftarrow \text{Gen}$
 - $\text{Dec}_{sk}(\text{Enc}_{pk}(m)) = m$



Fully Homomorphic Encryption

- FHE Scheme: (Gen, **Enc**, Dec, **Eval**)
 - $(pk, sk) \leftarrow \text{Gen}$
 - $\text{Dec}_{sk}(\text{Eval}_{pk}(f, \text{Enc}_{pk}(m))) = f(m)$



Lattice-based Encryption

- secret key: $sk = s \in \mathbb{Z}^n$
- $Enc_s(m) = (A, As + m\Delta + e) = (A, b)$
 - $A \leftarrow$ random matrix
 - $e \leftarrow$ random noise ($|e| < \Delta/2$)
- $Dec_s(A, b)$:
 - $c = (b - As) / \Delta = m + e/\Delta$
 - output $round(c) = m$

Additive Homomorphism

- $C_0 = \text{Enc}_s(m_0) = (A_0, A_0s + m_0\Delta + e_0)$
- $C_1 = \text{Enc}_s(m_1) = (A_1, A_1s + m_1\Delta + e_1)$
- $C_0 + C_1 = ((A_0 + A_1), (A_0 + A_1)s + (m_0 + m_1)\Delta + (e_0 + e_1))$
- If $|e_i| < \Delta/4$, then
 - $|e_0 + e_1| < \Delta/2$
 - $C_0 + C_1$ is a valid encryption of $m_0 + m_1$
- ... but $C_0 + C_1 + C_2$ may not decrypt correctly

Gentry bootstrapping

- Technique to reduce the amount of error in a lattice-based ciphertext:
 - Assume $(\text{Gen}, \text{Enc}, \text{Dec}, \text{Eval})$ supports the evaluation of $f_c(\text{sk}) = \text{Dec}_{\text{sk}}(c)$
 - Publish $\text{ek} = \text{Enc}(\text{sk})$ as part of public key
 - Output $c' = \text{Eval}(f_c, \text{ek})$
- Computing Dec homomorphically clears the noise of c
- Requires Enc to be “circularly secure”

Bootstrapping and Circular Security

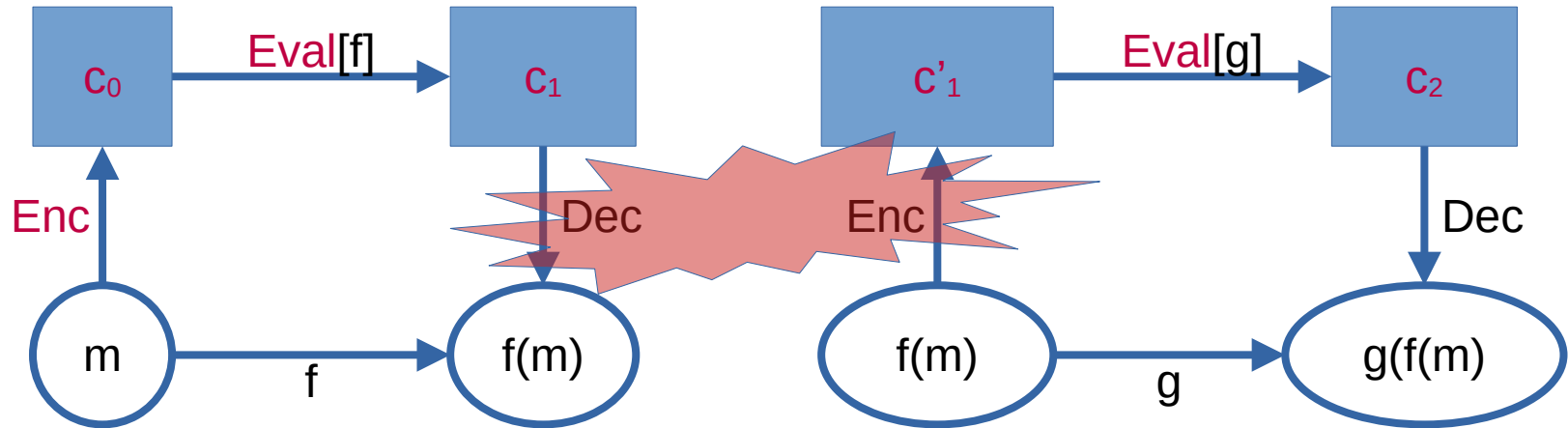
- All known FHE schemes use bootstrapping
- Bootstrapping requires circular security
- Questions:
 - Can you transform any given FHE scheme into a new one FHE' which is circular secure? ([...,AP16,GKW17,HK17]: no, at least for non-homomorphic encryption)
 - Can you prove circular security of FHE from standard LWE assumption?
- Still an open problem ...

Composition

- Perform two computations in sequence:
 - Input : $m \rightarrow f(m) \rightarrow g(f(m))$: output
 - $c_0 = \text{Enc}(m)$
 - $c_1 = \text{Eval}(f, c_0)$
 - $c_2 = \text{Eval}(g, c_1)$
- Question: $\text{Dec}(c_2) = g(f(m))$?
- Answer: not necessarily

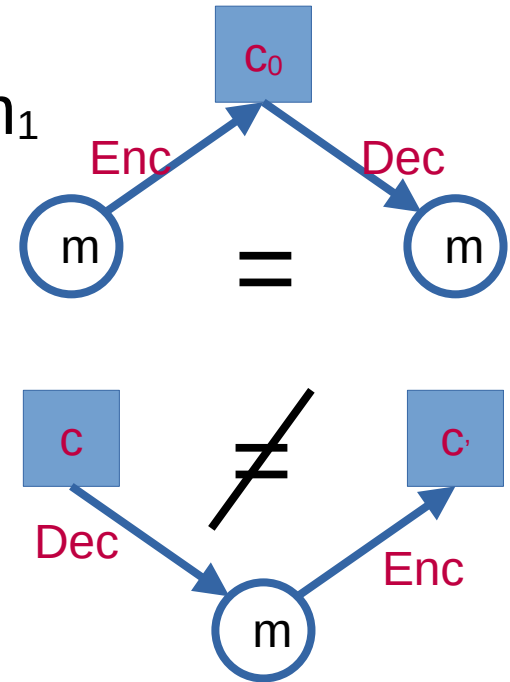
The Problem with Composition

- $c_0 = \text{Enc}(m)$
- $c_1 = \text{Eval}(f, c_0)$
- $c_2 = \text{Eval}(g, c_1)$



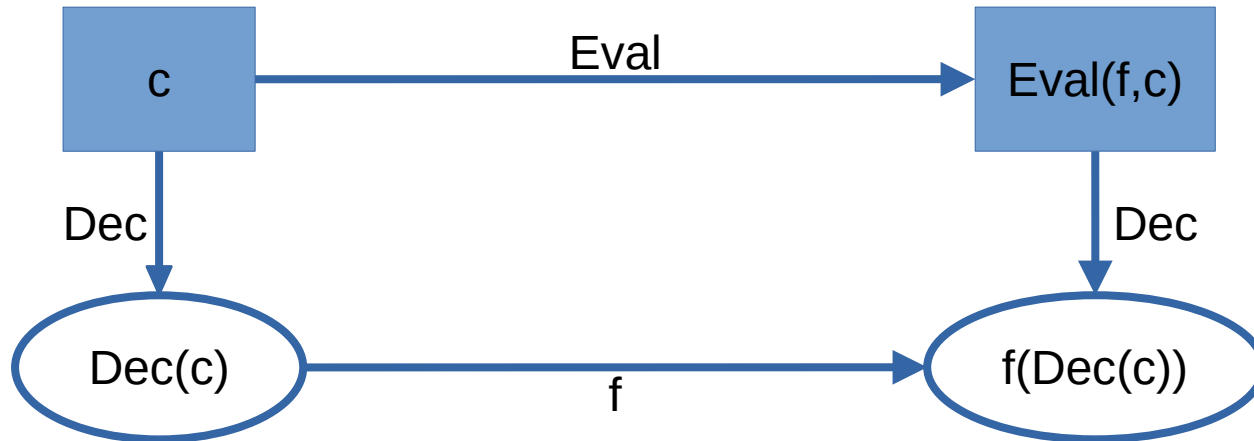
Correctness of Encryption

- Property 1: $c_1 = \text{Enc}(m_1)$
- Property 2: $\text{Dec}(c_1) = m_1$
- Correctness of encryption: $\text{Dec}(\text{Enc}(m_1)) = m_1$
 - (Property 1) implies (Property 2)
- But the converse may not be true
 - (Property 2) does not imply (Property 1)
 - $\text{Enc}(\text{Dec}(c_1))$ is not in general equal to c_1



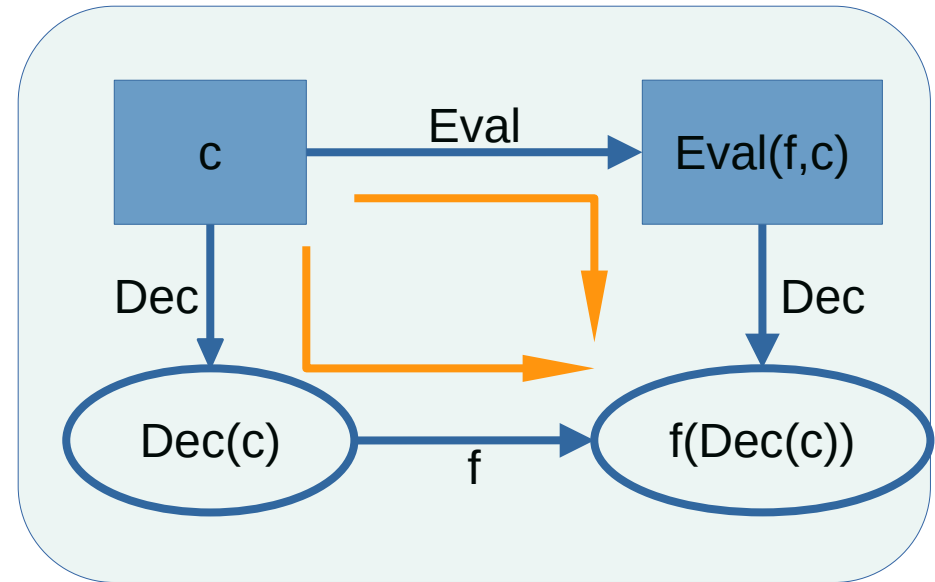
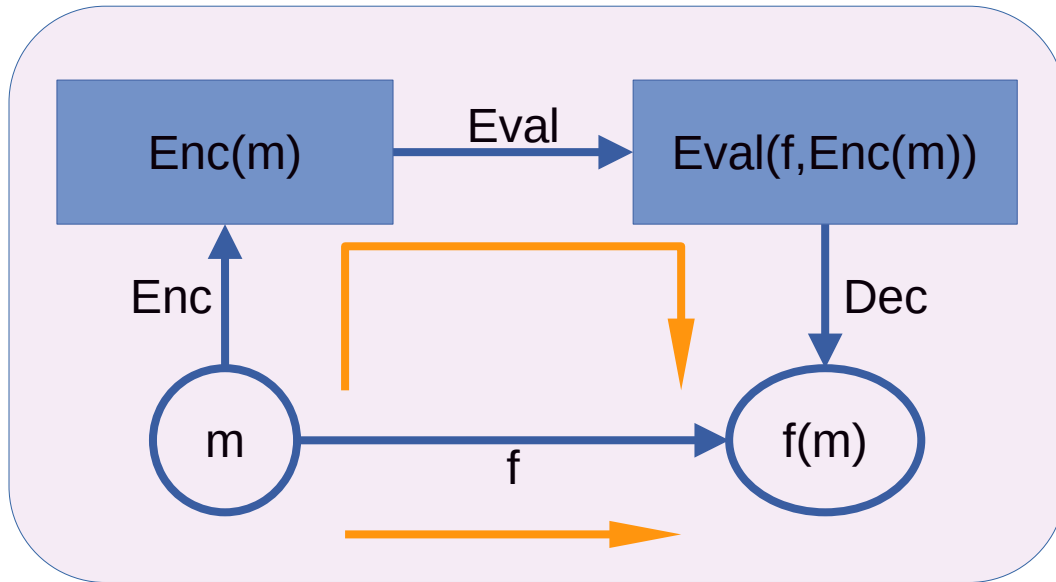
Fully Composable HE

- FHE Scheme: (Gen,Enc,Dec,Eval)
- A new correctness property:
 - $\text{Dec}(\text{Eval}(f,c)) = f(\text{Dec}(c))$



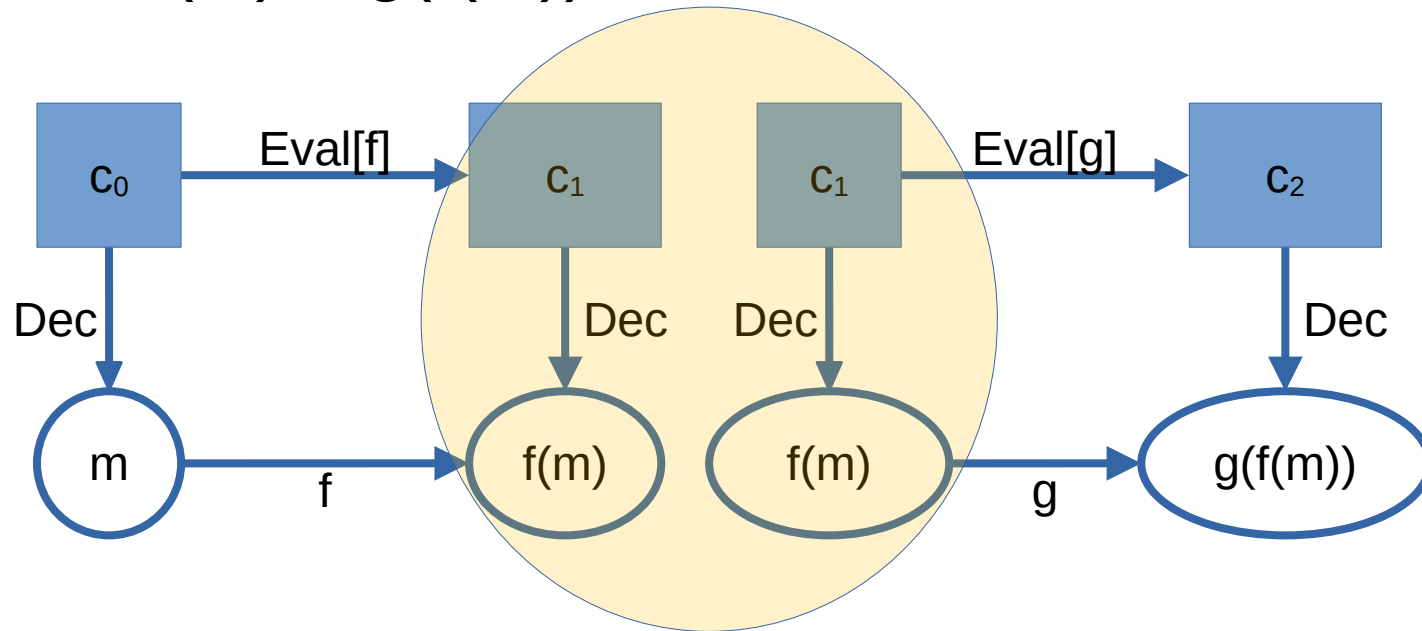
Comparing the two definitions

- FHE: $\text{Dec}(\text{Eval}(f, \text{Enc}(m))) = f(m)$
- CFHE: $\text{Dec}(\text{Eval}(f, c)) = f(\text{Dec}(c))$



Composition

- $c_0 = \text{Enc}(m)$; $c_1 = \text{Eval}(f, c_0)$; $c_2 = \text{Eval}(g, c_1)$
- $\text{Dec}(c_2) = g(f(m))$!



CFHE implies FHE

- FHE: $\text{Dec}(\text{Eval}(f, \text{Enc}(m))) = f(m)$
- CFHE: $\text{Dec}(\text{Eval}(f, c)) = f(\text{Dec}(c))$
- Let $c = \text{Enc}(m)$. Then

$$\text{Dec}(\text{Eval}(f, \text{Enc}(m)))$$

$$= \text{Dec}(\text{Eval}(f, c))$$

[definition of c]

$$= f(\text{Dec}(c))$$

[CFHE]

$$= f(\text{Dec}(\text{Enc}(m)))$$

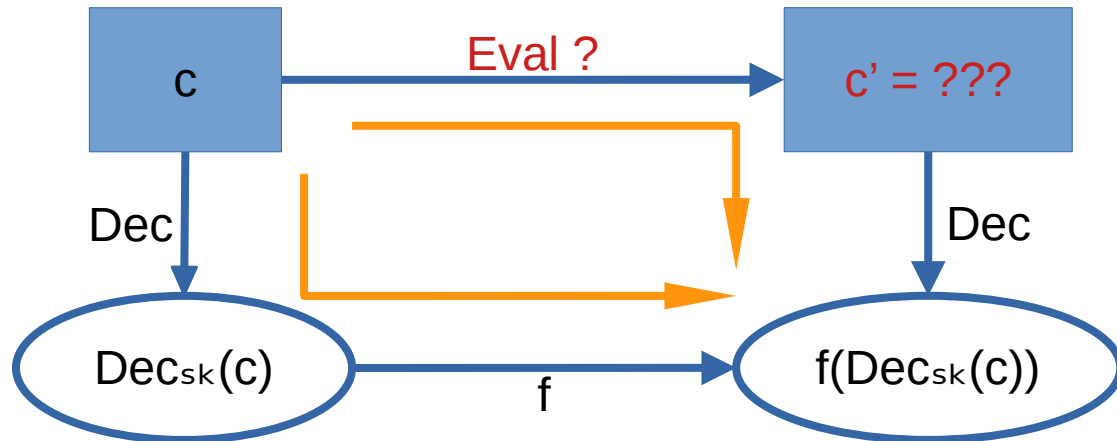
[definition of c]

$$= f(m)$$

[correctness of decryption]

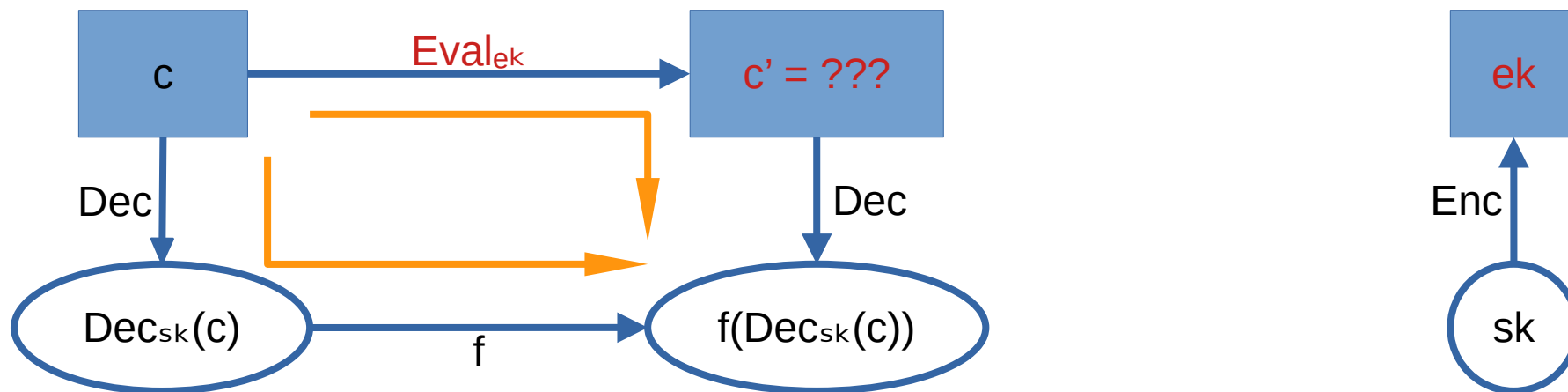
Bootstrapping revisited

- **Theorem:** FHE+circular security $\Rightarrow$ CFHE
 - Given FHE(Gen,Enc,Dec,Eval)
 - Build CFHE(Gen,Enc,Dec,Eval)
 - Goal: $\text{Dec}_{\text{sk}}(c') = f(\text{Dec}_{\text{sk}}(c))$



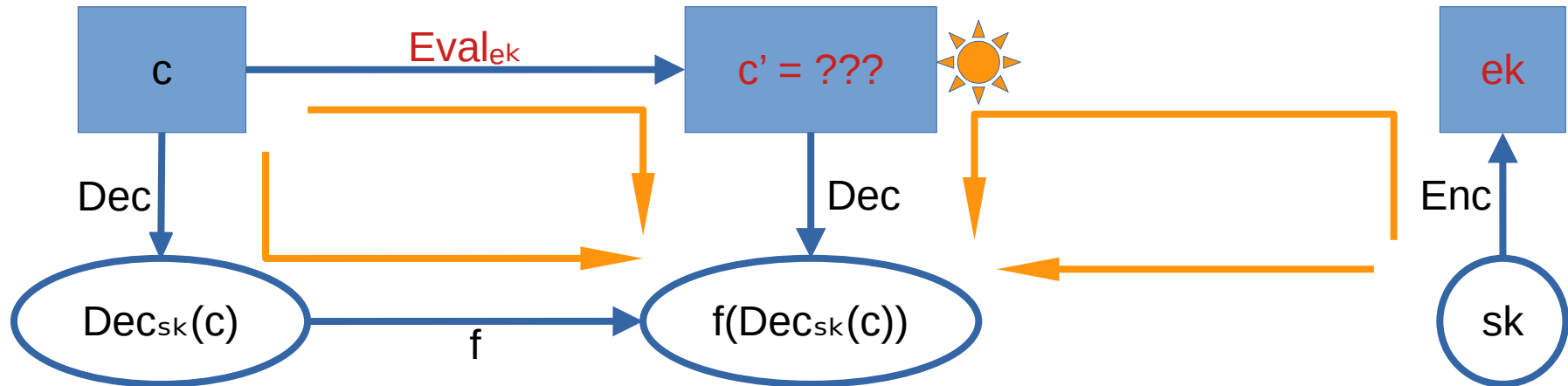
Bootstrapping revisited

- **Theorem:** FHE+circular security $\Rightarrow$ CFHE
 - **Gen:** $ek = \text{Enc}(sk)$



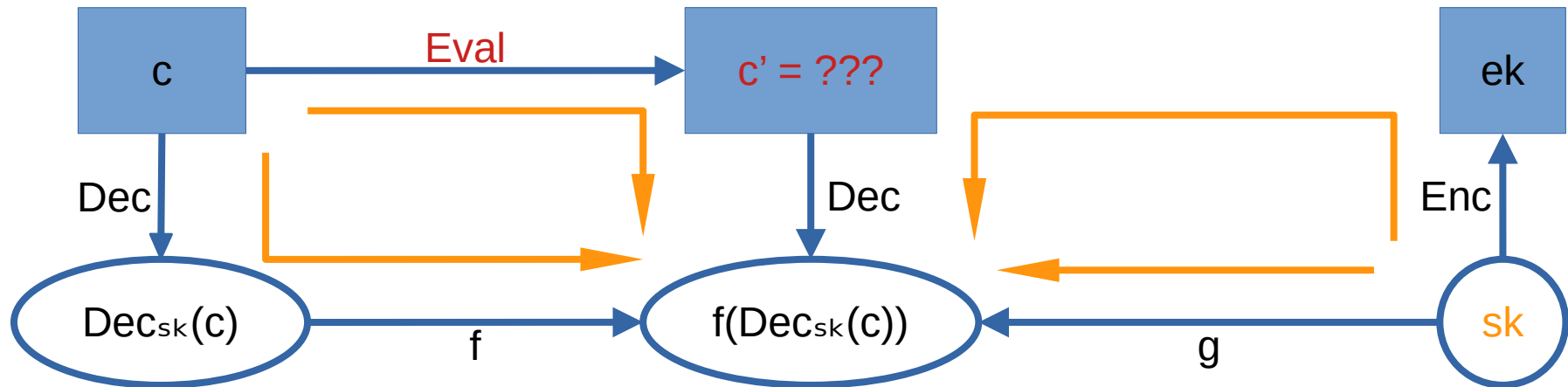
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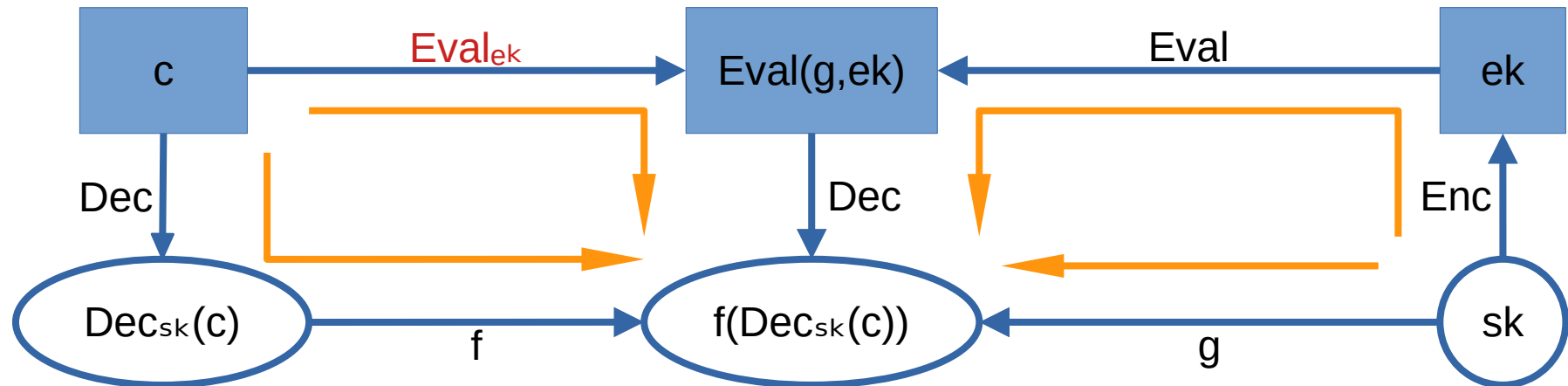
Bootstrapping revisited

- **Theorem:** FHE+circular security $\Rightarrow$ CFHE
 - Gen: $ek = \text{Enc}(sk)$
 - $g(sk) = f(\text{Dec}_{sk}(c))$



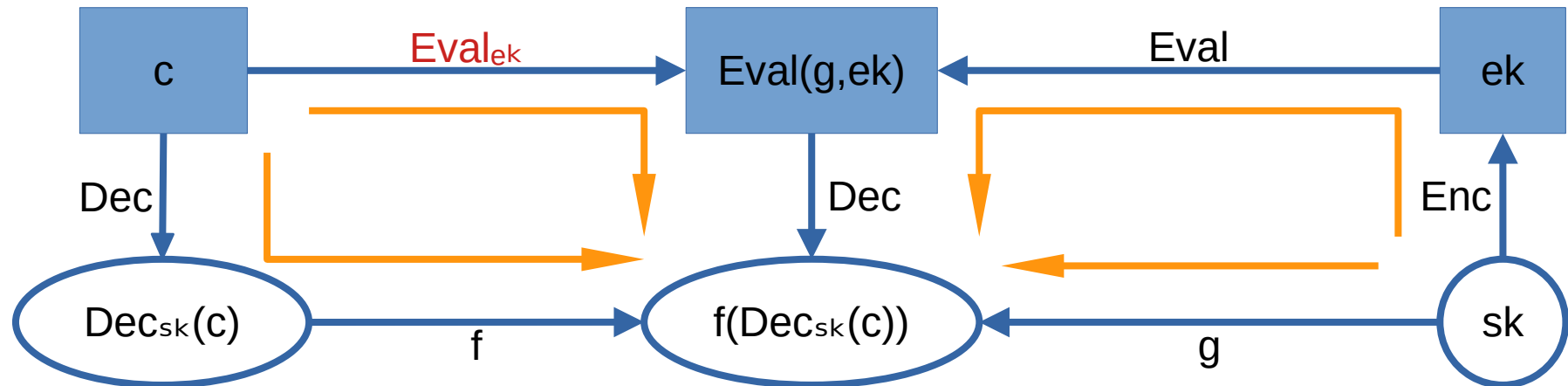
Bootstrapping revisited

- **Theorem:** FHE+circular security $\Rightarrow$ CFHE
 - **Gen:** $ek = \text{Enc}(sk)$
 - $g(sk) = f(\text{Dec}_{sk}(c))$
 - $\text{Eval}_{ek}(f, c) = c' = \text{Eval}(g, ek)$



Bootstrapping revisited

- **Theorem:** FHE+circular security $\Rightarrow$ CFHE
- Theorem/proof makes no mention of
 - ciphertexts being noisy
 - need to reduce ciphertext noise to keep computing

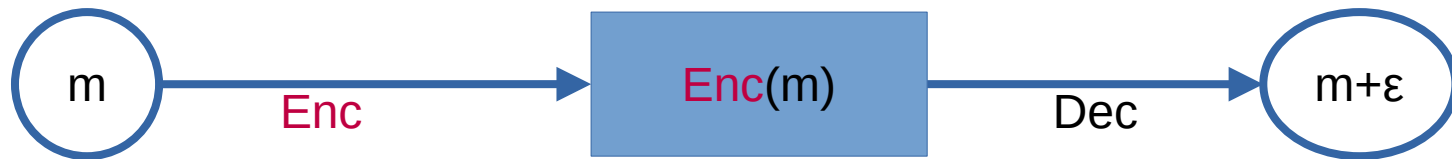


CFHE without circular security?

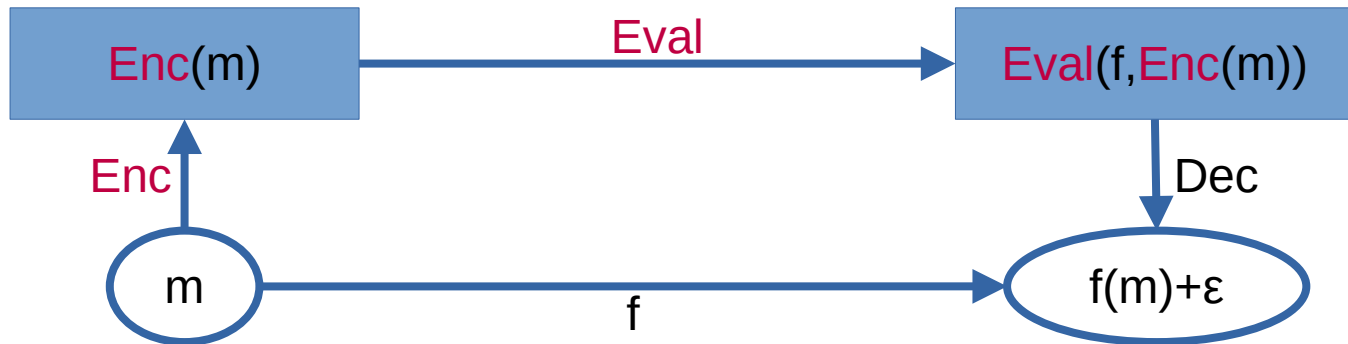
- Question: Does FHE implies CFHE?
 - Given FHE scheme $(\text{Gen}, \text{Enc}, \text{Dec}, \text{Eval})$
 - Build new CFHE scheme $(\text{Gen}', \text{Enc}', \text{Dec}', \text{Eval}')$
- Note: question does not involve circular security!
- Alternative 1: no such transformation is possible
 - give a black box separation between FHE and CFHE)
- Alternative 2: $(\text{FHE} \Rightarrow \text{CFHE}) \Rightarrow \text{circular security}$
 - prove that any such transformation implies circular security

Approximate Encryption Scheme

- $\text{Dec}_{\text{sk}}(\text{Enc}_{\text{pk}}(m)) = m + \varepsilon$



- $\text{Dec}_{\text{sk}}(\text{Eval}_{\text{pk}}(f, \text{Enc}_{\text{pk}}(m))) = f(m) + \varepsilon$



Approximate Encryption (CKKS)

- Standard: $\text{Enc}_s(m) = (A, As+m\Delta+e)$
- Approximate:
 - $\text{Enc}_s(m) = (A, As+m+e)$
 - $\text{Dec}_s(A,b) = b - As = m+e \approx m$
- Removing Δ leads to substantial efficiency improvement due to slower error growth

Security of (Approx.) LWE encryption

- Regev 2005: Learning With Errors (LWE)
 - $(A, As+e) \approx (A, b)$ with b uniformly random
- Ciphertexts leak no information about m :
 - $\text{Enc}_s(m) = (A, As+e+m\Delta) \approx (A, b)$
 - $\text{Enc}_s(m) = (A, As+e+m) \approx (A, b)$
- Is this enough to securely use LWE encryption?

Insecurity of CKKS17 [LM21]

- Compute statistics on encrypted dataset $x[1, \dots, n]$
 - $\text{Enc}(x[1]), \text{Enc}(x[2]), \dots, \text{Enc}(x[n])$
 - Compute $C = C[1] + C[2] + \dots + C[n]$
 - $\text{Decrypt}(C) = x' \approx x = x[1] + x[2] + \dots + x[n]$
- Assume the exact value of x becomes known
- Enough to recover the secret key sk from $(x - x')$!
 - This is bad: it may allow to decrypt other, sensitive datasets
- Approximate FHE requires a new, stronger security definition than ciphertext indistinguishability [LM21, LMSS22]

Too much for one talk

- “On the security of Homomorphic Encryption on Approximate Numbers”,
Li, **M.**, **Eurocrypt 2021**
- “Securing Approximate Homomorphic Encryption using Differential Privacy”,
Li, **M.**, Schultz, Sorrell, **Crypto 2022** (to appear)

Approximate vs Exact Encryption

- Approximate encryption $\Rightarrow$ Exact encryption
 - $\text{Dec}(\text{Enc}(m)) = m + \varepsilon$ for some $|\varepsilon| < \delta$
 - $\text{Enc}'(m) = \text{Enc}(2\delta m)$
 - $\text{Dec}'(c) = \text{round}(\text{Dec}(c)/(2\delta))$
- Question: can approximate FHE be more efficient than FHE ?
 - Theoretical formulation of efficiency gap ?